ADNAN MALIK

Data Science | Analytics | Business Intelligence | Data Engineering
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Open to Relocation: UAE | Saudi Arabia | Qatar

PROFESSIONAL SUMMARY

Data Science & Analytics Professional with 5+ years of experience designing cloud-based data solutions, automated ETL pipelines, and interactive Power BI dashboards that turn raw data into decision-ready insight. Experienced across SQL, Python, Power BI, Azure, GCP, and AWS, with a strong track record of building scalable data architectures, integrating API-driven datasets, developing reporting and warehousing layers, and delivering KPI reporting for operational and executive stakeholders. Backed by an MSc in Data Science & Analytics and hands-on projects in predictive modelling, machine learning, statistical analysis, and data visualisation, bringing a blend of business intelligence, cloud data engineering, and advanced analytics.

CORE COMPETENCIES

Data Engineering & ETL Pipelines
Cloud Data Solutions & Warehousing
SQL, Data Modelling & Database Management
Business Intelligence & Reporting
Data Analysis & Visualisation
Python Automation & API Integration
Data Quality, KPI Reporting & Governance
Stakeholder Management & Requirements Gathering
Machine Learning & Predictive Analytics

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Prolog
BI & Visualisation: Power BI, Power BI Service, Tableau, Qlik Sense
Cloud & Data Platforms:

- **AWS:** CloudWatch, EC2, S3, Lambda, EMR, Glue Crawlers, Athena, SES
- **GCP:** Cloud Run functions, GCS, BigQuery
- **Azure:** Azure SQL Server, Databricks, Azure Data Factory

Other Tools & Technologies: Advanced Excel, ETL Pipelines, API Integration, Jupyter Notebook, GitHub

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PROFESSIONAL EXPERIENCE

Data & Systems Analyst
Calex UK (Henry Ford Academy), United Kingdom | **May 2023 – Present**
Technologies: SQL, Python, Power BI, Power BI Service, Qlik, API Integration, GCP (Cloud Run Functions, GCS, BigQuery), Azure (Azure SQL Server, Databricks, Azure Data Factory)

- Designed and implemented a GCP-based data solution using Cloud Run Functions, GCS, and BigQuery to ingest raw data from external APIs and transform it into structured, analytics-ready datasets.
- Built automated ETL pipelines and reporting-layer datasets to support data warehousing, dashboarding, and downstream reporting requirements.
- Developed Qlik dashboards connected to GCP BigQuery to monitor training progress and delegate performance, with group-based access controls to ensure users could view only relevant data.
- Developed a parallel Azure-based data architecture using Azure Data Factory, Databricks, and Azure SQL Server to replicate ingestion, transformation, and reporting workflows in support of business continuity and future cloud transition plans.

- Built and maintained interactive Power BI dashboards and executive performance reports for operational and management teams.
- Developed Power BI reports connected to Azure SQL Server and configured automated refresh schedules through Power BI Service to deliver up-to-date reporting without manual intervention.
- Implemented job-role-level row-level security in Power BI dashboards to control external access and ensure each user could view only the records relevant to their role.
- Designed dynamic dashboards to monitor Training Standards and delegate performance, using drill-through, slicers, filters, and KPI indicators to support detailed performance analysis.
- Applied SQL, Python, data modelling, transformation, and data cleansing techniques to improve data quality, streamline reporting workflows, and increase reporting efficiency.

Key Achievements:

- Designed and maintained dual-cloud data solutions across GCP and Azure, supporting business continuity and cloud transition strategy.
- Delivered secure BI reporting across Qlik and Power BI, using group-based and job-role-level access controls to support controlled access to training and performance data.
- Reduced manual reporting effort through API ingestion automation and scheduled refresh implementation.

Analyst → Senior Analyst
LatentView Analytics, India | May 2019 – August 2021
Technologies: SQL, Python, AWS (Lambda, S3, EMR, Glue Crawlers, Athena, CloudWatch, EC2, SES), Hadoop, Advanced Excel

- Delivered AWS-based ETL and data warehousing solutions to support business intelligence reporting and downstream data consumption.
- Built and maintained ETL pipelines using Lambda, S3, EMR, Glue Crawlers, and Athena to ingest, transform, and prepare clickstream data for reporting and warehousing layers.
- Created structured datasets and data models for dashboard development, KPI reporting, and reporting-layer data availability.
- Implemented automated monitoring and validation checks to improve pipeline reliability, strengthen data quality, and reduce manual effort.
- Worked with stakeholders to gather reporting requirements and support KPI-focused reporting deliverables.
- Supported the GDPR-compliant migration of Hadoop-based data infrastructure to AWS, helping improve scalability, resilience, and compliance.

Key Achievements:

- Reduced manual monitoring workload by approximately 2 hours per day through automation and workflow improvements.
- Improved reporting accuracy and pipeline reliability through automated validation and monitoring processes.

PROJECTS

Machine Learning for Solubility Prediction of Chemical Compounds

Tools: Python, Machine Learning, Predictive Modelling, Cross-Validation, Data Visualisation, Dissertation Research

- Conducted a dissertation project focused on predicting the **solubility of chemical compounds** using machine learning under conditions of **experimental uncertainty and noisy data**.
- Built and evaluated **14 machine learning models** across five modelling approaches: linear regression methods, latent-variable methods, ensemble methods, kernel-based models, and neural networks.
- Applied a structured modelling workflow involving **data checks, feature screening, train-test splitting, 10-fold cross-validation, hyperparameter tuning, and consensus-based prediction**.

- Evaluated model performance using **RMSE, Pearson's r^2, %LogS±0.7, and %LogS±1.0**, reflecting both statistical accuracy and tolerance for unavoidable experimental error in solubility data.
- **Result / Outcome:** prediction performance improved through the use of relevant descriptors, evaluation methods that accounted for experimental error, and a consensus of top-performing models. Additional gains were achieved through refinement of the feature set to reduce redundancy and retain the most informative variables.

Image Classification using Convolutional Neural Networks

Tools: Python, PyTorch, CNNs, Computer Vision, TinyImageNet30

- Built and fine-tuned a **convolutional neural network** for image classification using **TinyImageNet30**, covering dataset preparation, model development, evaluation, and testing.
- Applied a practical deep-learning workflow including **single-batch overfitting checks**, full-dataset training, regularisation, and fine-tuning.
- Analysed learned visual features by inspecting **filters and feature maps** from the first convolutional layer of a pre-trained network.
- Explored the effect of increasing training data, augmentation, and **dropout** on model behaviour and generalisation.
- **Result / Outcome:** the coursework showed that the initial model overfit the training data and performed poorly on validation data, while **data augmentation** and **dropout** improved validation performance and helped reduce overfitting.

Image Caption Generation using PyTorch

Tools: Python, PyTorch, Deep Learning, Computer Vision, NLP

- Built an **image caption generation system** in Python and PyTorch using **COCO_5029**, a subset of the COCO dataset containing **5,029 images** with multiple reference captions.
- Developed an end-to-end workflow covering **image feature extraction, text preprocessing, vocabulary construction, decoder training, prediction generation, and metric-based evaluation**.
- Implemented an encoder-decoder style captioning pipeline in which image features from a pre-trained **ResNet-152** encoder were passed to an **RNN-based decoder** for sequence generation.
- Evaluated generated captions using **BLEU score** and **cosine similarity** to compare predicted captions with reference captions.
- **Result / Outcome:** the coursework showed that **BLEU** and **cosine similarity** can behave very differently for the same poor prediction, highlighting the value of using more than one evaluation metric when assessing caption quality.

MLP Classification with PyTorch

Tools: Python, PyTorch, Deep Learning, Neural Networks

- Implemented a **multi-layer perceptron (MLP)** in PyTorch to classify handwritten digits from the **MNIST dataset**.
- Covered the full workflow including **image loading, tensor conversion, neural-network construction using torch.nn, and model training with cross-entropy loss and stochastic gradient descent**.
- Trained and compared **one-layer and two-layer neural-network architectures** to understand how model structure affects classification behaviour.
- Repeated training to reduce loss and improve classification performance.
- **Learning / Outcome:** the project provided practical exposure to implementing, training, and comparing neural-network classifiers in PyTorch.

Mushroom Classification: Edible or Poisonous

Tools: R, Decision Trees, Random Forest, Classification, Cross-Validation

- Developed **tree-based classification models** to predict whether mushrooms were **edible or poisonous** using features such as cap shape, cap surface, cap colour, odour, and height.

- Built and tuned both **Decision Tree** and **Random Forest** models using **10-fold cross-validation** to identify the best-performing classifiers.
- Compared model structure and feature importance to determine which attributes contributed most strongly to classification performance.
- Assessed classifier reliability in the context of a high-stakes classification task where incorrect edible predictions could have serious consequences.
- **Result / Outcome:** both models classified the large majority of mushrooms correctly, and **height** was identified as a weak feature with limited predictive contribution. The analysis also highlighted that even a small false-positive rate could carry serious real-world risk in this use case.

Analysis of NHS Dental Statistics

Tools: Python, Jupyter Notebook, Data Preparation, Statistical Analysis, Data Visualisation

- Prepared a linked analytical dataset to study **NHS dental statistics in England** for the period **31/03/2018 to 31/03/2019** at **CCG level**, using **five raw datasets** covering dentists, patients, deprivation, and CCG mapping.
- Cleaned, merged, and quality-checked the raw data to produce a final dataset covering **191 CCGs**, then used Python notebooks for analysis and visualisation.
- Analysed treatment patterns by age, examined the effect of **deprivation**, and projected potential **dentist shortages in 2031** based on workforce ageing.
- Studied the relationship between treated patient numbers and deprivation index across England's Clinical Commissioning Groups.
- **Result / Outcome:** the analysis found that the highest number of treated patients came from **Age-Bands 8, 9 and 10**, that younger children aged **0–2** formed a small share of treated patients, that treated-patient levels across **Age-Bands 8–16** were broadly similar, and that deprivation showed a slight positive relationship with treated-patient counts. It also projected a decline in dentist numbers by **2031**, with **NHS Berkshire West** showing the highest expected shortage.

Statistical Analysis of "Age at First Kill" of Serial Killers

Tools: R, RStudio, Statistical Analysis, Hypothesis Testing, Data Visualisation

- Conducted an **R-based statistical analysis** to examine how the **motive behind murder** may affect the career timeline of serial killers.
- Focused on **age at first kill** as the main variable of interest, while also examining related timeline measures such as age at last kill and active duration.
- Performed data cleaning, hypothesis testing, and supporting statistical analysis through R-based scripts and visual outputs.
- Tested a previously reported benchmark that the average age at first kill was **27**.
- **Result / Finding:** the analysis found that the average age at first kill was **lower than 27** for gang-related killers and **higher than 27** for those who killed to escape and those categorised as **Angels of Death**, indicating meaningful variation across motive groups.

Analysing Effects of Gender Distribution in School Performance

Tools: Python, Statistical Analysis, Public Dataset Analysis

- Analysed schools in England using government school-performance data to assess whether **gender distribution and school type** were associated with performance outcomes.
- Evaluated school performance using **OFSTED ratings** together with pupil performance measures in **reading, writing, and mathematics**.
- Examined both the number and proportion of boys and girls in schools to understand how school composition related to outcomes.
- Structured the work around public-sector data analysis and performance interpretation using statistical methods.

- **Result / Finding:** the analysis found that mixed schools were the majority, girls-only and boys-only schools were split roughly evenly, girls-only schools appeared to perform better than boys-only schools, and schools with a **higher percentage of girls** tended to perform better overall.

Mobile Phones Price-Range Prediction

Tools: Python, Machine Learning, Classification, Exploratory Data Analysis

- Developed a **machine learning classification project** to predict **mobile phone price range** using technical and hardware-related product features.
- Worked with separate **training and test datasets**, using the training data to analyse feature relationships and train a classification model, and the test data for evaluation.
- Analysed a **21-column feature set** containing device attributes such as battery power, memory, processor-related features, screen dimensions, connectivity, and RAM.
- Combined exploratory data analysis with supervised model development to build a structured classification workflow for price-range prediction.
- **Result / Outcome:** the project established a supervised-learning workflow for translating device specifications into price-range classification using structured product-feature data.

Workplace Reasoning Knowledge Base in Prolog

Tools: Prolog, Knowledge Representation, Rule-Based Reasoning, Logic Inference

- Built a **Prolog knowledge base** to model workplace hierarchy, employee behaviour, and rule-based consequences in an organisational setting centred on **Aramco** and **IMALCO**.
- Defined facts and inference rules for roles such as CEO, manager, developer, and accountant, together with workplace conditions such as **sleeping at work**, **pending work**, and management preferences.
- Structured the logic to derive outcomes such as **promotion, dismissal, movement to another branch, salary cuts, and transfer to another company**.
- Designed rule chains to connect employee actions and workplace context to inferred organisational consequences.
- **Result / Outcome:** the knowledge base successfully inferred who could be **promoted, dismissed, or moved**, achieving **inference depth 7** with a relatively compact rule set. The evaluation also identified limitations including incomplete domain coverage and the absence of dynamic fact updates after inferred role changes.

Walmart Sales & Customer Analysis

Tools: Tableau, Excel, Data Visualisation

- Built a **Tableau dashboard** to analyse Walmart sales and customer data over the period from **2012 to 2016**.
- Explored sales performance across products and customer segments to understand purchasing behaviour and segment-level trends.
- Designed interactive visual reports to support comparison across customer groups and commercial performance measures.
- Used dashboard-based analysis to support sales review and customer-segmentation insight generation.
- **Result / Outcome:** the dashboard provided useful insight into how different **customer segments can be targeted**.

Hotel Revenue & Customer Analysis

Tools: Power BI, Excel, Data Visualisation

- Developed an interactive **Power BI dashboard** to analyse hotel revenue and customer trends using booking and operational data from **January 2018 to September 2020**.

- Examined key metrics including **overall revenue, year-wise and month-wise revenue, reservations, cancellations, guest volumes, market-segment behaviour, discount behaviour, meal revenue, and customer nationality trends**.
- Used interactive reporting features to study hotel performance, strengths, weaknesses, and patterns relevant to business improvement.
- Built dashboard views to support business review, operational planning, and performance monitoring.
- **Result / Outcome:** the analysis supported the identification of **patterns and areas for improvement**, future planning, and comparison of **more profitable versus less profitable market segments**.

IoT-Based Electric Vehicle Fault Detection & Battery Monitoring

Tools: MATLAB, Simulink, Arduino IDE, NodeMCU, ThingSpeak, IFTTT, IoT, Linear Regression, Battery Monitoring, Sensors

- Developed an IoT-based capstone project to detect electric-vehicle battery faults in real time and transmit fault data to the cloud for monitoring and response.
- Designed the solution around battery health and charging-state monitoring, using SOC and SOH estimation methods together with linear-fit / linear-regression-based processing to support fault detection and prediction.
- Built a hardware-based data acquisition setup using Arduino, NodeMCU, and sensors to capture battery voltage, current, and temperature data and transmit it to the cloud platform.
- Implemented real-time cloud monitoring through ThingSpeak, enabling data visualisation, signal processing, and automated SMS alerts when battery conditions such as low SOC or SOH crossed defined thresholds.
- Tested the solution on a prototype EV-style setup using a lithium-ion battery and brushless DC motor to validate end-to-end data capture, transmission, monitoring, and alerting.

Result / Outcome: the project demonstrated an autonomous fault-detection and battery-monitoring system that successfully sent real-time battery data to the cloud, supported linear-fit-based battery-health analysis, and triggered driver notifications when unusual battery conditions were detected.

Masarif — Personal Expense Tracker (React PWA)

Tools: React 19, Vite, Chart.js, JavaScript, CSS, PWA (Workbox), Cloudflare Pages, Data Visualisation

- Built a **privacy-first, offline-capable personal expense dashboard** that runs entirely in the browser with no account or server required, deployed as a PWA on Cloudflare Pages.
- Implemented **CSV and Excel bank statement ingestion** with smart column auto-detection, multi-file merge, and robust deduplication logic to handle overlapping statements from multiple accounts.
- Developed an interactive **analytics dashboard** featuring KPI cards, daily flow line charts, income vs. spend bar charts, expense breakdown doughnuts, a date-range slider, and period binning across salary cycle, calendar month, quarter, and year views.
- Built a **Transaction Explorer drawer** with multi-pill filters, payee search, per-row exclusion toggles, bulk exclude/include, paginated table, and CSV export; and a Payee Lookup panel with per-payee stats and monthly trend charts.
- **Result / Outcome:** a fully client-side, installable expense tracker with no data leaving the user's device, demonstrating end-to-end frontend engineering across data parsing, interactive visualisation, responsive mobile layout, and PWA deployment.

EDUCATION

University of Leeds, United Kingdom
MSc Data Science & Analytics | **2021 –2022**
Grade: Pass with Distinction
Focus: Data Analysis, Statistical Analysis, Machine Learning, Artificial Intelligence, Deep Learning
Dissertation: *Machine Learning for Solubility Prediction of Chemical Compounds*

SRM Institute of Science & Technology, Chennai, India
B.Tech Electrical and Electronics Engineering | 2015 – 2019
Grade: Pass with Distinction
Focus: Electrical, Electronics, Mathematics, Programming Fundamentals
Capstone Project: *IoT-Based Real-Time Fault Detection and Transmission for Electric Vehicles*

TRAINING & CERTIFICATIONS

- **SQL** — Stanford Online | May 2019
- **Advanced SQL for Data Scientists** — LinkedIn Learning | Aug 2021
- **Applied Machine Learning: Foundations** — LinkedIn Learning | Aug 2021
- **Machine Learning Advanced Certification Training** — Simplilearn | May 2020
- **Data Scientist with Python Track** — DataCamp | Nov 2020
- **Big Data Hadoop and Spark Developer** — Simplilearn | Jan 2020
- **Analytics Bootcamp 101** — LatentView Analytics | Jun 2019
- **LatentView Data Services Foundation** — Great Learning x LatentView | Feb 2020

AWARDS & RECOGNITION

- **Reward of Excellence –** Henry Ford Academy (2023)
- **Encore Award –** LatentView Analytics (2020)
- **Certificate of Commendation –** Royal Statistical Society, Leeds-Bradford Local Group (2022): Awarded for demonstrating exceptional content, understanding, and presentation quality during the MSc Data Science & Analytics poster session at the University of Leeds.
- **Certificate of Merit –** SRM University, Department of Physics & Nanotechnology (2016): Awarded Best Project/Working Model at TechKnow-2016 for the Dark Sensing Circuit project, adjudged by an expert technical committee.
- **Certificate of Appreciation –** SRM University, Career Development Centre (2016): Recognised for outstanding performance in the Technical Presentation & Personality Development Course, Department of Electrical and Electronics Engineering.

LANGUAGES

English (Fluent, IELTS 7.5 – C1) | Urdu (Native) | Hindi (Native) | Arabic (Conversational)